

TE Curve Verification Pipeline Familywise ONNX Report - harmonic_regression

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates exported ONNX models from the dataset input-mode retraining program. Each dataset/input-mode section uses dataset-matched held-out test curves and lists the exact model artifacts loaded from `models/`.

The report is diagnostic and family-specific. It does not replace an official multi-index model-promotion decision.

Output Artifacts

- output directory: `output/validation_checks/track2_familywise_onnx_report/harmonic_regression/2026-07-08-09-24-54__track2_harmonic_regression_familywise_onnx_report` ;
- summary YAML: `output/validation_checks/track2_familywise_onnx_report/harmonic_regression/2026-07-08-09-24-54__track2_harmonic_regression_familywise_onnx_report/track2_familywise_onnx_report_summary.yaml` ;
- model inventory CSV: `output/validation_checks/track2_familywise_onnx_report/harmonic_regression/2026-07-08-09-24-54__track2_harmonic_regression_familywise_onnx_report/model_inventory.csv` ;
- per-curve metrics CSV:
`output/validation_checks/track2_familywise_onnx_report/harmonic_regression/2026-07-08-09-24-54__track2_harmonic_regression_familywise_onnx_report/per_curve_metrics.csv` .

Simplified Dataset + Setpoints

- dataset: `simplified_dataset` ;
- input mode: `setpoints` ;
- evaluated family: `harmonic_regression` ;
- dataset root: `data/simplified_dataset` .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	<code>te_harmonic_regression_fw_simplified_setpoints</code>	<code>2026-07-07-22-54-59_te_harmonic_regression_fw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
backward	<code>te_harmonic_regression_bw_simplified_setpoints</code>	<code>2026-07-07-23-01-08_te_harmonic_regression_bw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
global	<code>te_harmonic_regression_global_simplified_setpoints</code>	<code>2026-07-07-22-48-30_te_harmonic_regression_global_simplified_setpoints</code>	<code>simplified_curve_v1</code>

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	<code>models/simplified_dataset/setpoints/exported/harmonic_regression/forward/2026-07-07-22-54-59_te_harmonic_regression_fw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/harmonic_regression/forward/2026-07-07-22-54-59_te_harmonic_regression_fw_simplified_setpoints/python/harmonic_regression-epoch=033-val_mae=0.01699562.ckpt</code>
backward	<code>models/simplified_dataset/setpoints/exported/harmonic_regression/backward/2026-07-07-23-01-08_te_harmonic_regression_bw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/harmonic_regression/backward/2026-07-07-23-01-08_te_harmonic_regression_bw_simplified_setpoints/python/harmonic_regression-epoch=060-val_mae=0.01698885.ckpt</code>
global	<code>models/simplified_dataset/setpoints/exported/harmonic_regression/global/2026-07-07-22-48-30_te_harmonic_regression_global_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/harmonic_regression/global/2026-07-07-22-48-30_te_harmonic_regression_global_simplified_setpoints/python/harmonic_regression-epoch=063-val_mae=0.01699328.ckpt</code>

Aggregate Metrics

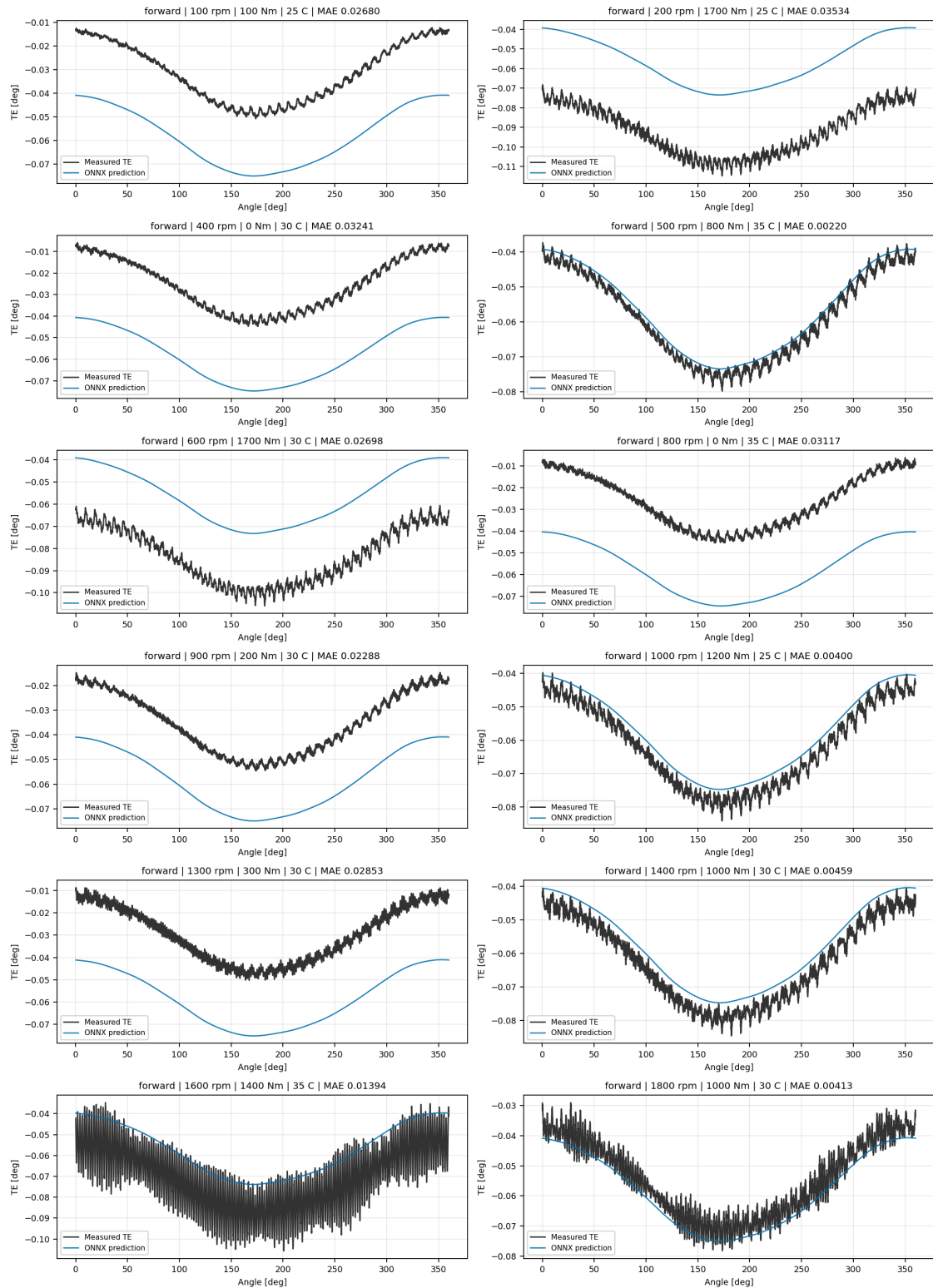
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.018064	0.018299	41.176	78.839
backward	97	0.017958	0.018234	41.314	85.556
global	194	0.018080	0.018336	41.405	81.780

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.001165	0.017990	0.001686	0.010423
backward	-0.000835	0.017779	0.001819	0.009625
global	0.000126	0.017954	0.001754	0.009988

Forward 12-Curve Page

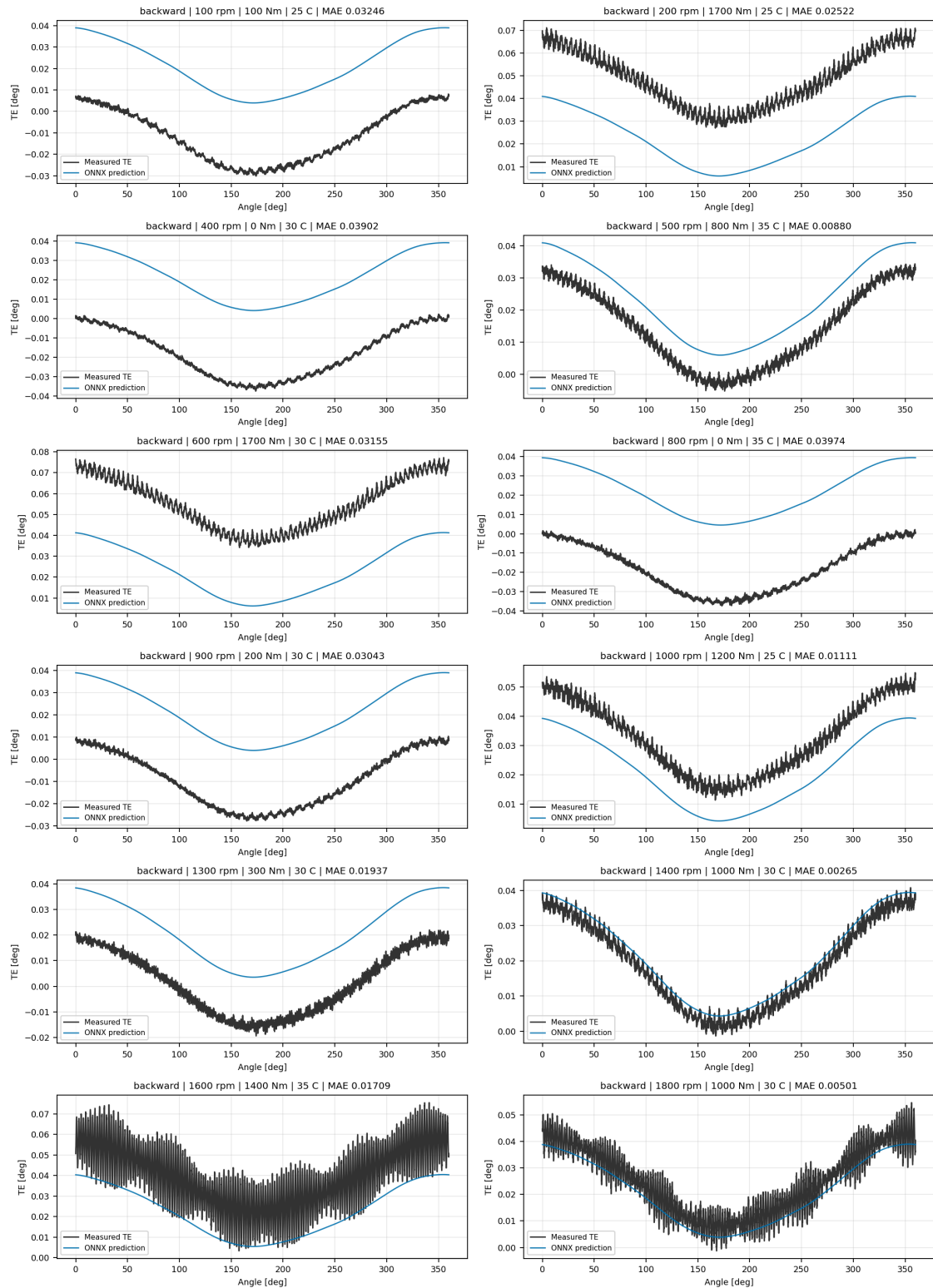
Simplified Dataset + Setpoints | harmonic_regression | forward



simplified_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

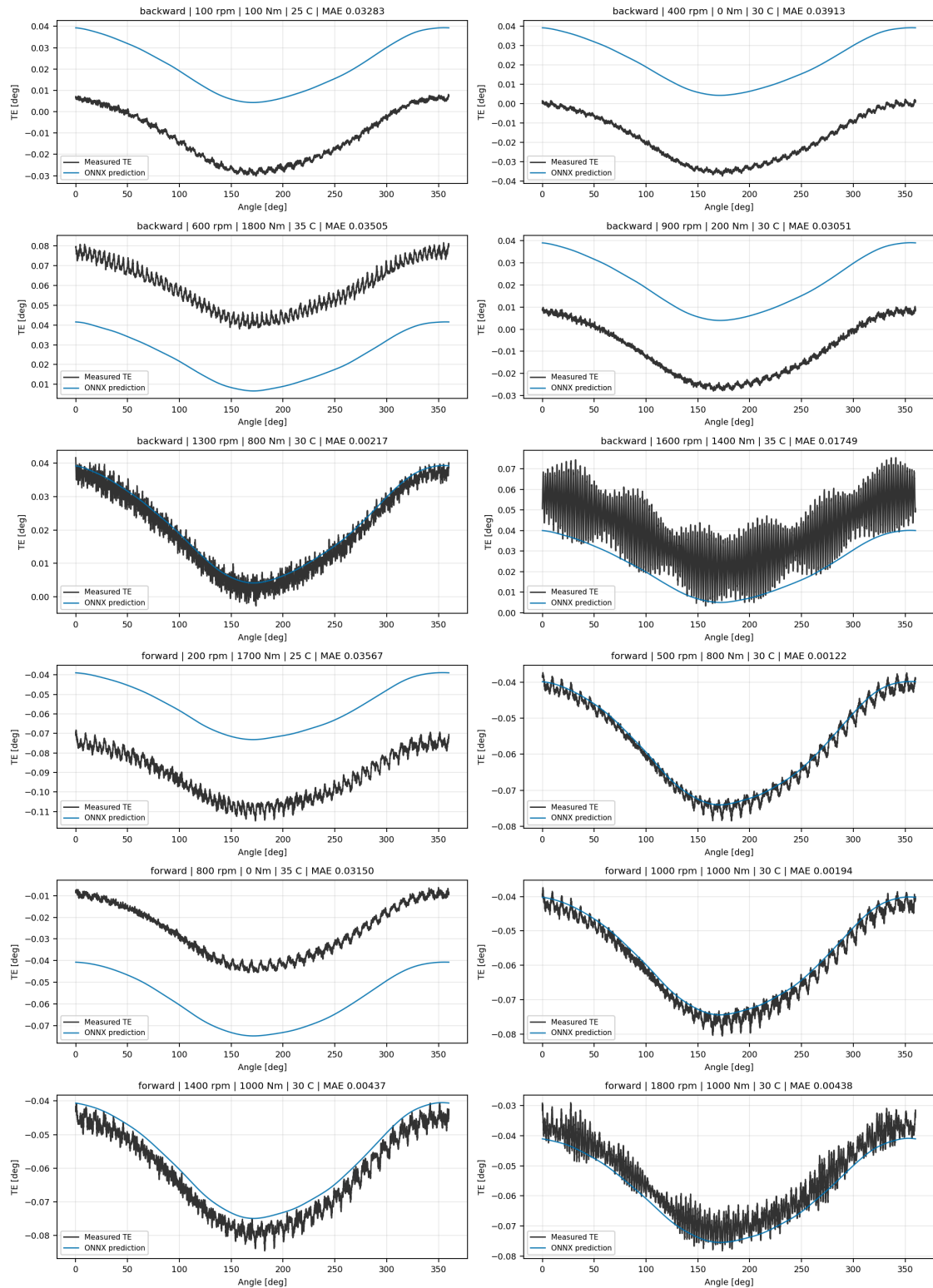
Simplified Dataset + Setpoints | harmonic_regression | backward



simplified_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Simplified Dataset + Setpoints | harmonic_regression | global



simplified_dataset_setpoints global 12-curve collage

Polished Dataset + Setpoints

- dataset: polished_dataset ;
- input mode: setpoints ;
- evaluated family: harmonic_regression ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_harmonic_regression_fw_polished_setpoints	2026-07-07-23-29-07_te_harmonic_regression_fw_polished_setpoints	polished_setpoint_curve_v1
backward	te_harmonic_regression_bw_polished_setpoints	2026-07-07-23-39-40_te_harmonic_regression_bw_polished_setpoints	polished_setpoint_curve_v1
global	te_harmonic_regression_global_polished_setpoints	2026-07-07-23-18-16_te_harmonic_regression_global_polished_setpoints	polished_setpoint_curve_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/setpoints/exported/harmonic_regression/forward/2026-07-07-23-29-07_te_harmonic_regression_fw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/harmonic_regression/forward/2026-07-07-23-29-07_te_harmonic_regression_fw_polished_setpoints/python/harmonic_regression-epoch=032-val_mae=0.01714993.ckpt
backward	models/polished_dataset/setpoints/exported/harmonic_regression/backward/2026-07-07-23-39-40_te_harmonic_regression_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/harmonic_regression/backward/2026-07-07-23-39-40_te_harmonic_regression_bw_polished_setpoints/python/harmonic_regression-epoch=044-val_mae=0.01715066.ckpt
global	models/polished_dataset/setpoints/exported/harmonic_regression/global/2026-07-07-23-18-16_te_harmonic_regression_global_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/harmonic_regression/global/2026-07-07-23-18-16_te_harmonic_regression_global_polished_setpoints/python/harmonic_regression-epoch=038-val_mae=0.01714113.ckpt

Aggregate Metrics

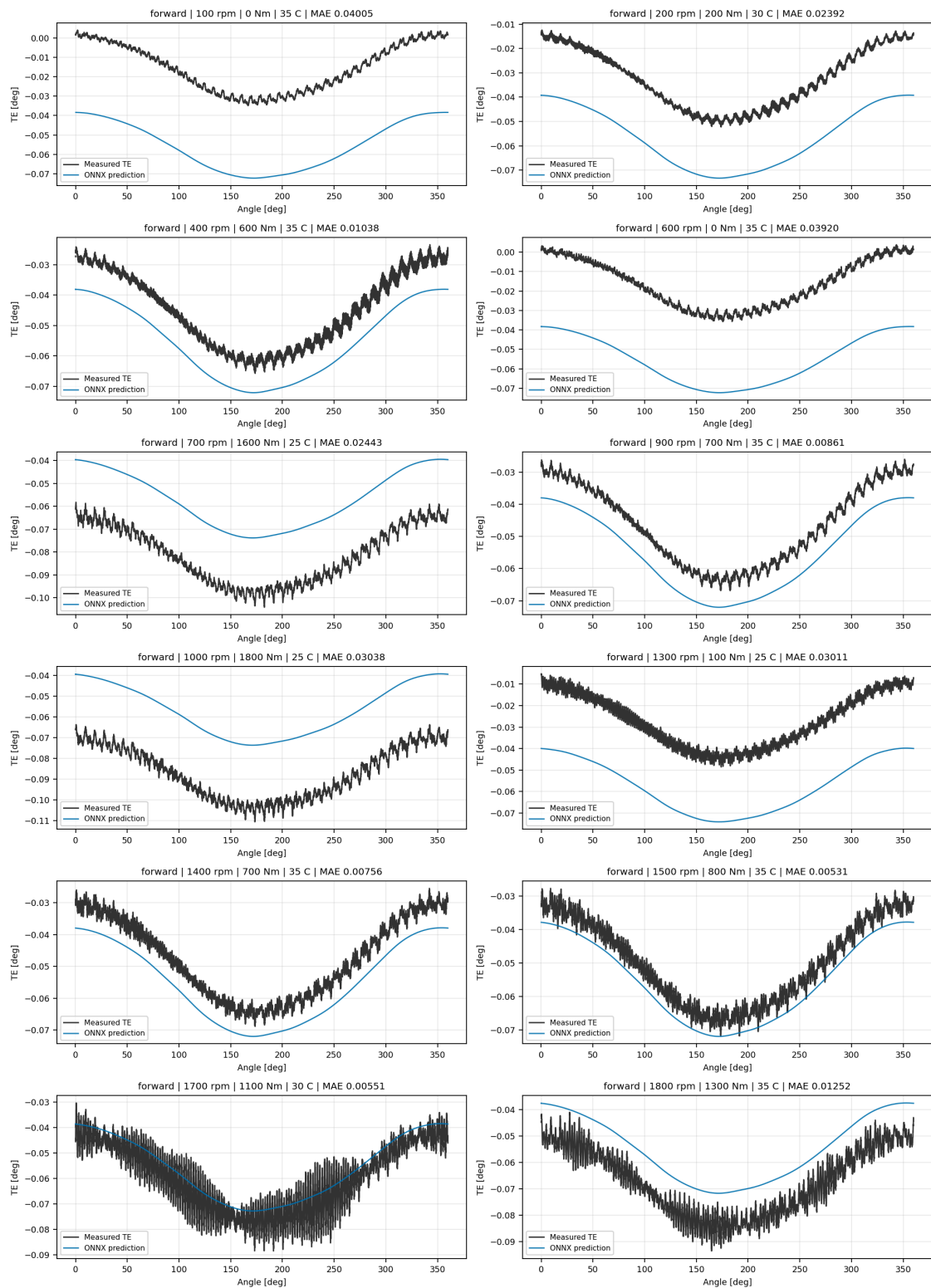
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.017220	0.017494	38.404	78.948
backward	94	0.017408	0.017823	37.675	77.694
global	194	0.017346	0.017688	38.151	77.969

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.002228	0.017014	0.001923	0.011496
backward	0.002071	0.017259	0.002351	0.011906
global	-0.000175	0.017164	0.002131	0.011713

Forward 12-Curve Page

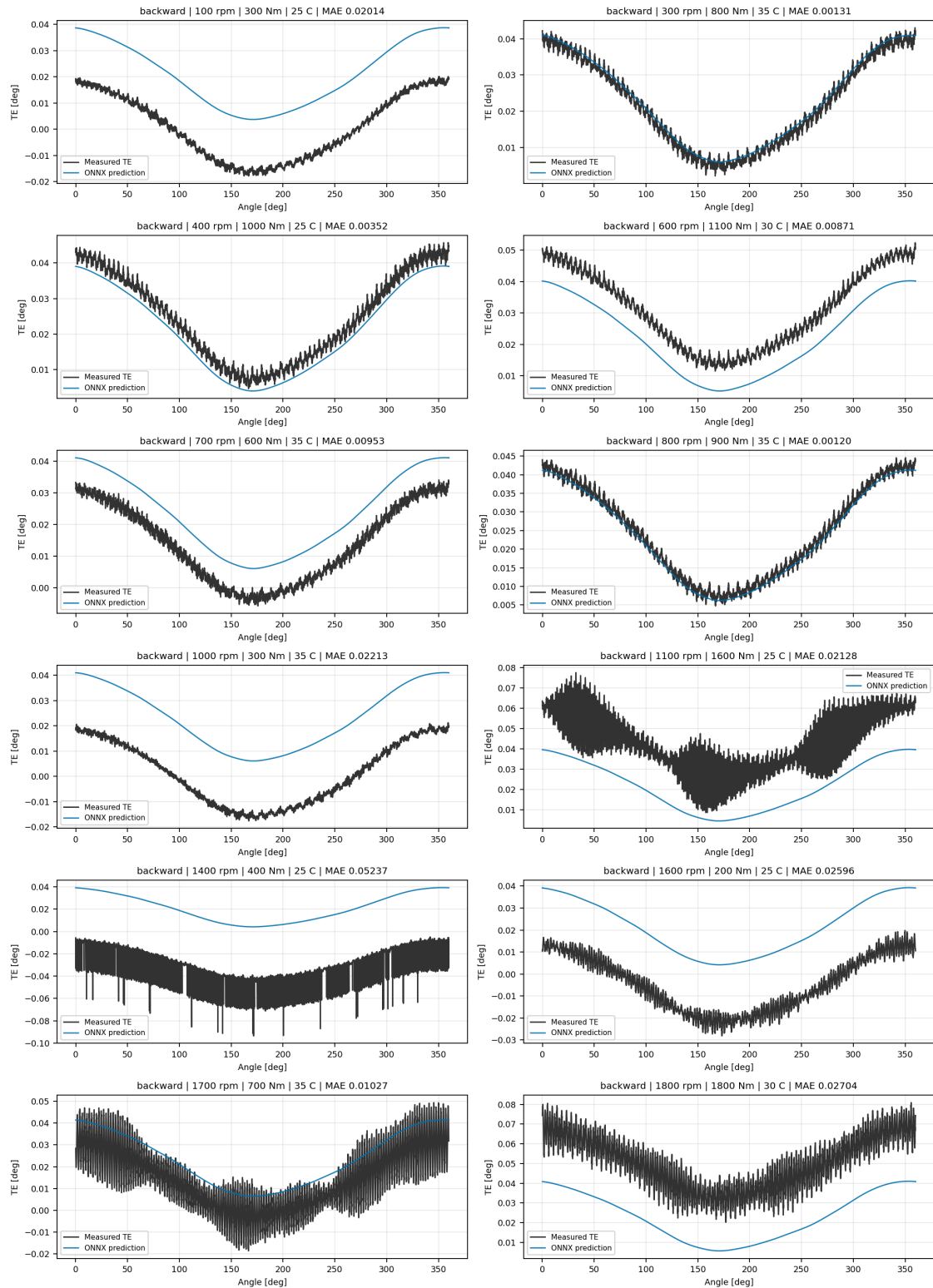
Polished Dataset + Setpoints | harmonic_regression | forward



polished_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

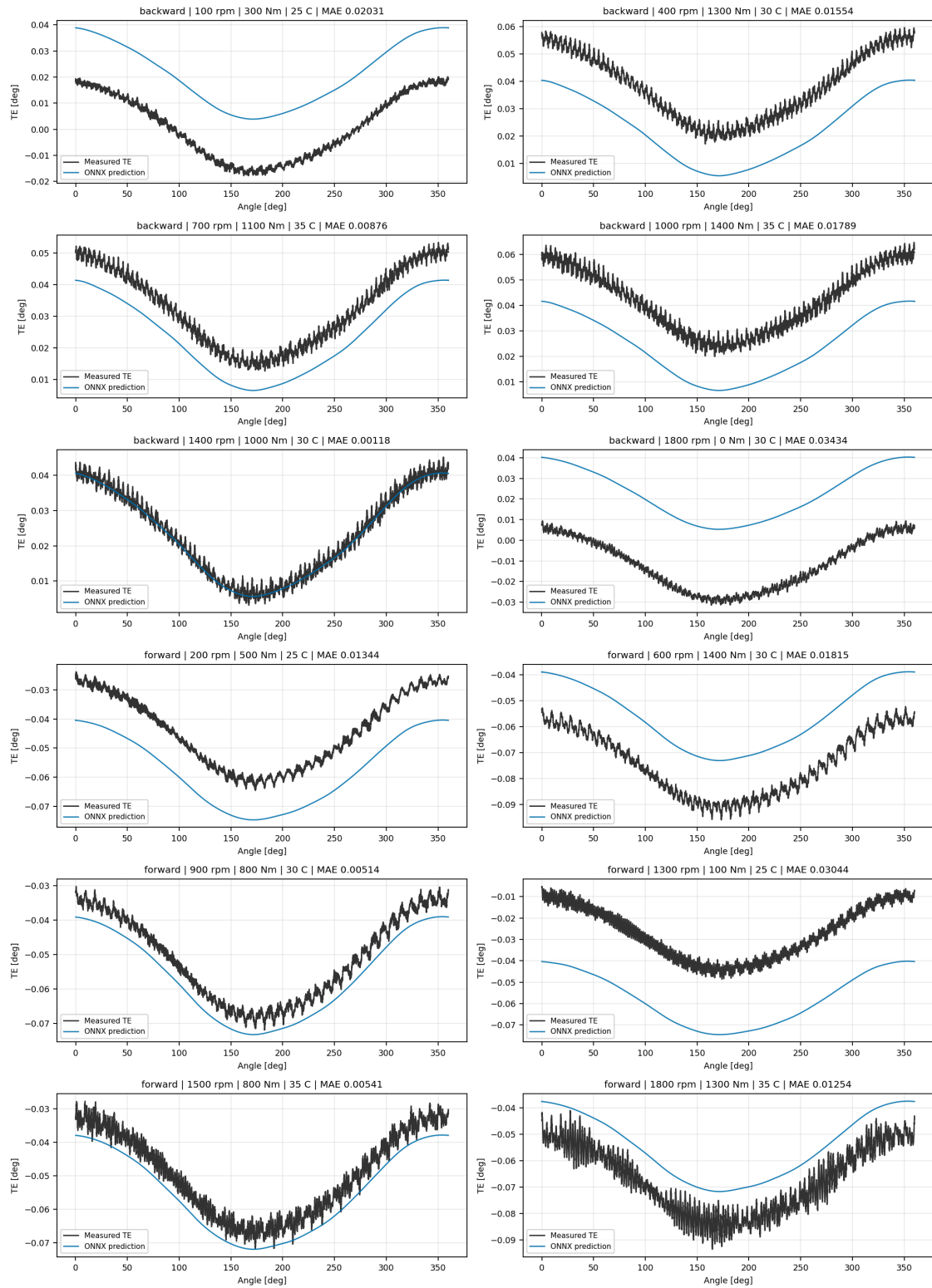
Polished Dataset + Setpoints | harmonic_regression | backward



polished_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Setpoints | harmonic_regression | global



polished_dataset_setpoints global 12-curve collage

Polished Dataset + Actual Values

- dataset: polished_dataset ;
- input mode: actual_values ;
- evaluated family: harmonic_regression ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_harmonic_regression_fw_polished_actual_values	2026-07-08-00-15-14_te_harmonic_regression_fw_polished_actual_values	polished_point_v1
backward	te_harmonic_regression_bw_polished_actual_values	2026-07-08-00-32-21_te_harmonic_regression_bw_polished_actual_values	polished_point_v1
global	te_harmonic_regression_global_polished_actual_values	2026-07-08-00-01-16_te_harmonic_regression_global_polished_actual_values	polished_point_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/actual_values/exported/harmonic_regression/forward/2026-07-08-00-15-14_te_harmonic_regression_fw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/harmonic_regression/forward/2026-07-08-00-15-14_te_harmonic_regression_fw_polished_actual_values/python/harmonic_regression-epoch=073-val_mae=0.00182314.ckpt
backward	models/polished_dataset/actual_values/exported/harmonic_regression/backward/2026-07-08-00-32-21_te_harmonic_regression_bw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/harmonic_regression/backward/2026-07-08-00-32-21_te_harmonic_regression_bw_polished_actual_values/python/harmonic_regression-epoch=054-val_mae=0.00182643.ckpt
global	models/polished_dataset/actual_values/exported/harmonic_regression/global/2026-07-08-00-01-16_te_harmonic_regression_global_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/harmonic_regression/global/2026-07-08-00-01-16_te_harmonic_regression_global_polished_actual_values/python/harmonic_regression-epoch=053-val_mae=0.00182331.ckpt

Aggregate Metrics

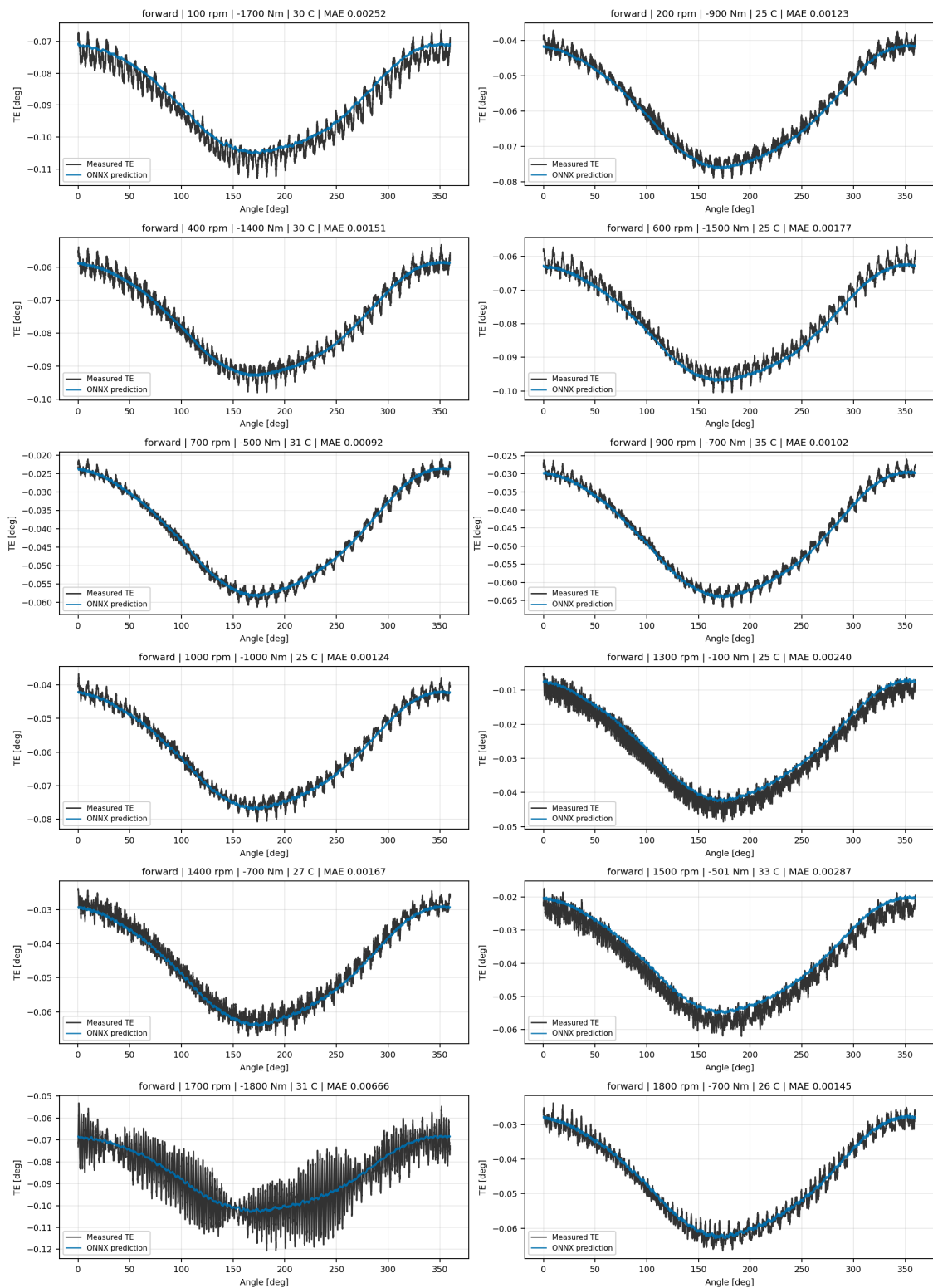
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.002355	0.002816	4.909	10.791
backward	94	0.002966	0.003511	5.369	10.974
global	194	0.002638	0.003138	5.101	10.859

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000051	0.001141	0.001938	0.010400
backward	-0.000118	0.001507	0.002402	0.010750
global	0.000031	0.001289	0.002162	0.010711

Forward 12-Curve Page

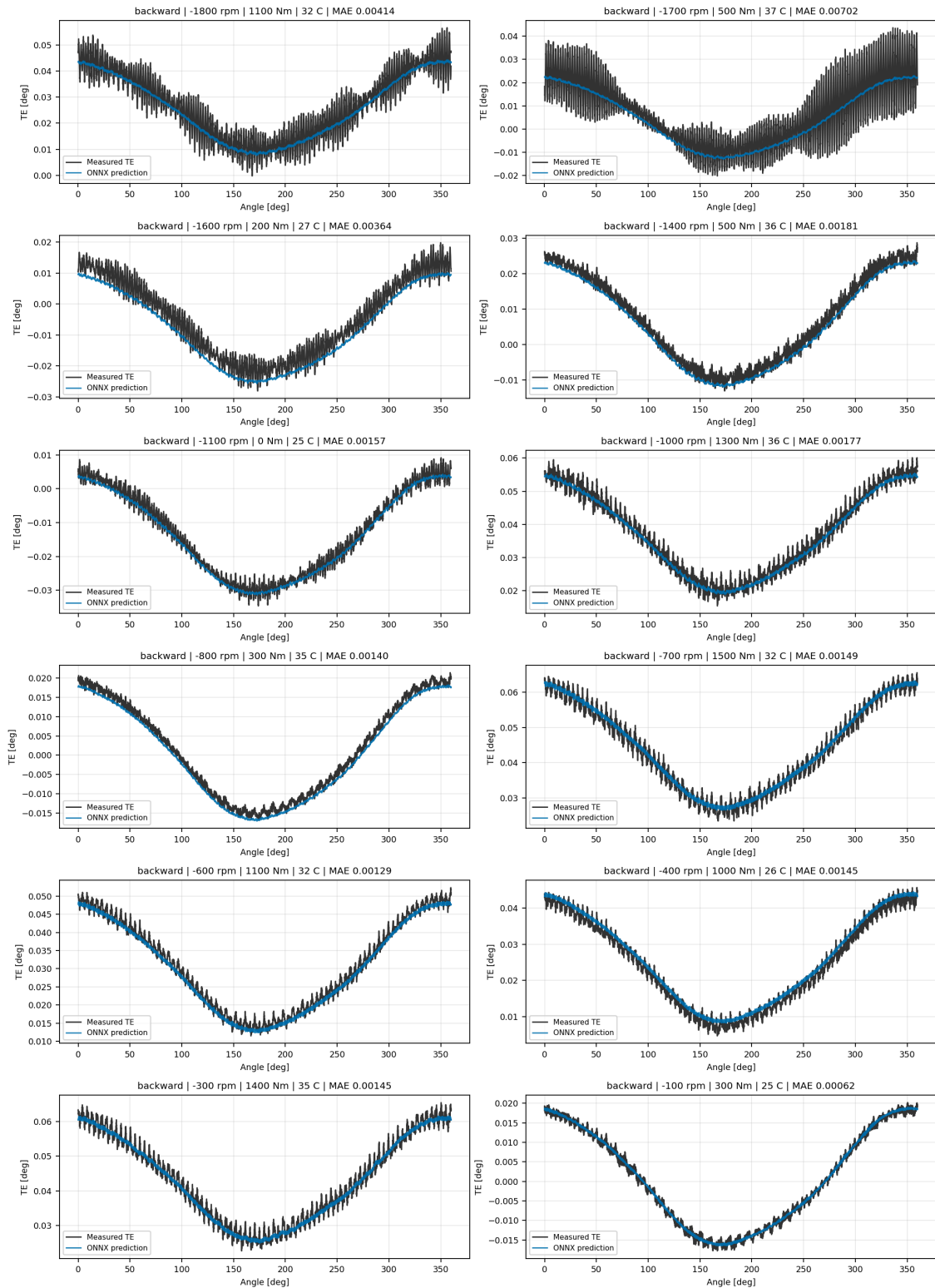
Polished Dataset + Actual Values | harmonic_regression | forward



polished_dataset_actual_values forward 12-curve collage

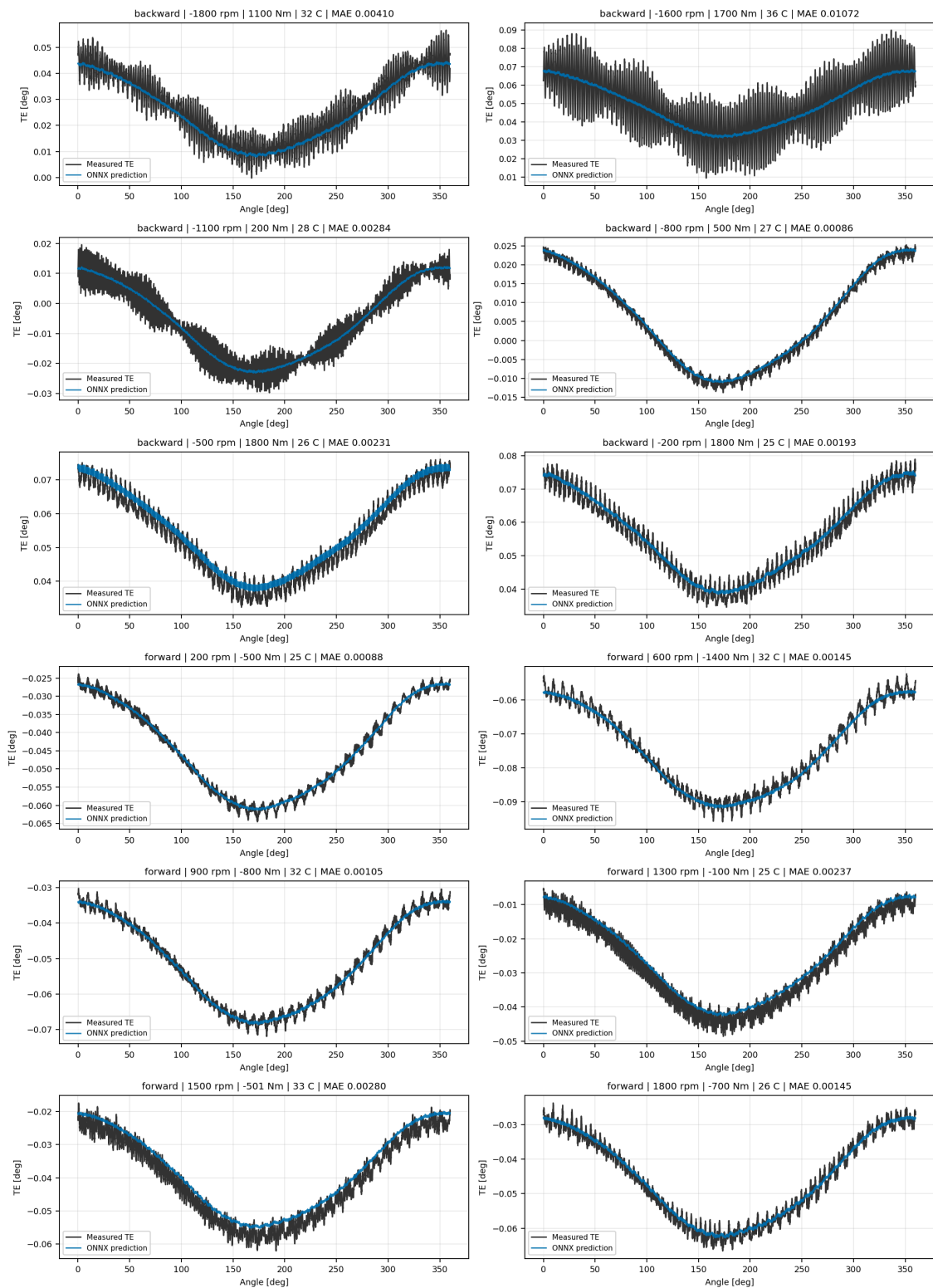
Backward 12-Curve Page

Polished Dataset + Actual Values | harmonic_regression | backward



polished_dataset_actual_values backward 12-curve collage

Polished Dataset + Actual Values | harmonic_regression | global



polished_dataset_actual_values global 12-curve collage